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OHC-2 Hardware Track

Our hardware, the results and conclusions

OpenFOAM Workshop 2026

Guimaraes, 1 July 2026



The BAW - who are we?

- Federal Waterways Engineering and Research Institute (BAW) in Germany
  - Scientific & technical support for Federal waterways and Shipping Administration (WSV)
  - Research in **hydraulic**, structural & geotechnical **engineering**
  - **Model testing** (physical & **numerical**) for flow, sediment & structure behaviour
  - Navigation design & waterway optimization
  - Environmental impact analysis & ecological planning



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Our cluster Amisia

- RHEL8.8 LUSTRE DDN Exascaler ES7990X;
  ~ 2500 TB usable, producer: Bull (former: Eviden)

- **146 compute nodes** (Bull Sequana XH3000)

  CPU:2 x AMD EPYC "Turin" 9755 (128 Cores each)
  RAM per node: 384 GB
  L3 Cache: 768 MB
  Max. TDP: 400W
  Total: **37 376 Cores** 112 TB RAM on the CPU nodes.

- **2 login nodes**
- **15 VIS-GPU Nodes** (Bull Sequana X450-A6)
- **5 AI GPU Nodes** (Bull Sequana X450-A6)



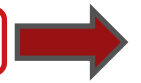
Our in-house HPC „Amisia“ during assembly

We use our HPCs for

- Hydrodynamic modelling with OpenFOAM, Telemac, DuMux, StarCCM+, Untrim
- Develop own models for ship dynamics in shallow waters,
- Machine learning research and development
- Data analysis with python etc.
- ....

- Our Hardware:
  Two clusters. ~ 57 000 CPU-Cores
  We keep the hardware for 5 years.

- OHC 2025: „Mosella“ CPUs: 102 x CPU: 2 AMD EPYC „Genoa“ 9654 (96 Cores each)
  OHC 2026: „Amisia“ CPUs: 146 x CPU: 2 AMD EPYC „Turin“ 9755 (128 Cores each)



OHC-2 2026

OHC Hardware track – our tests

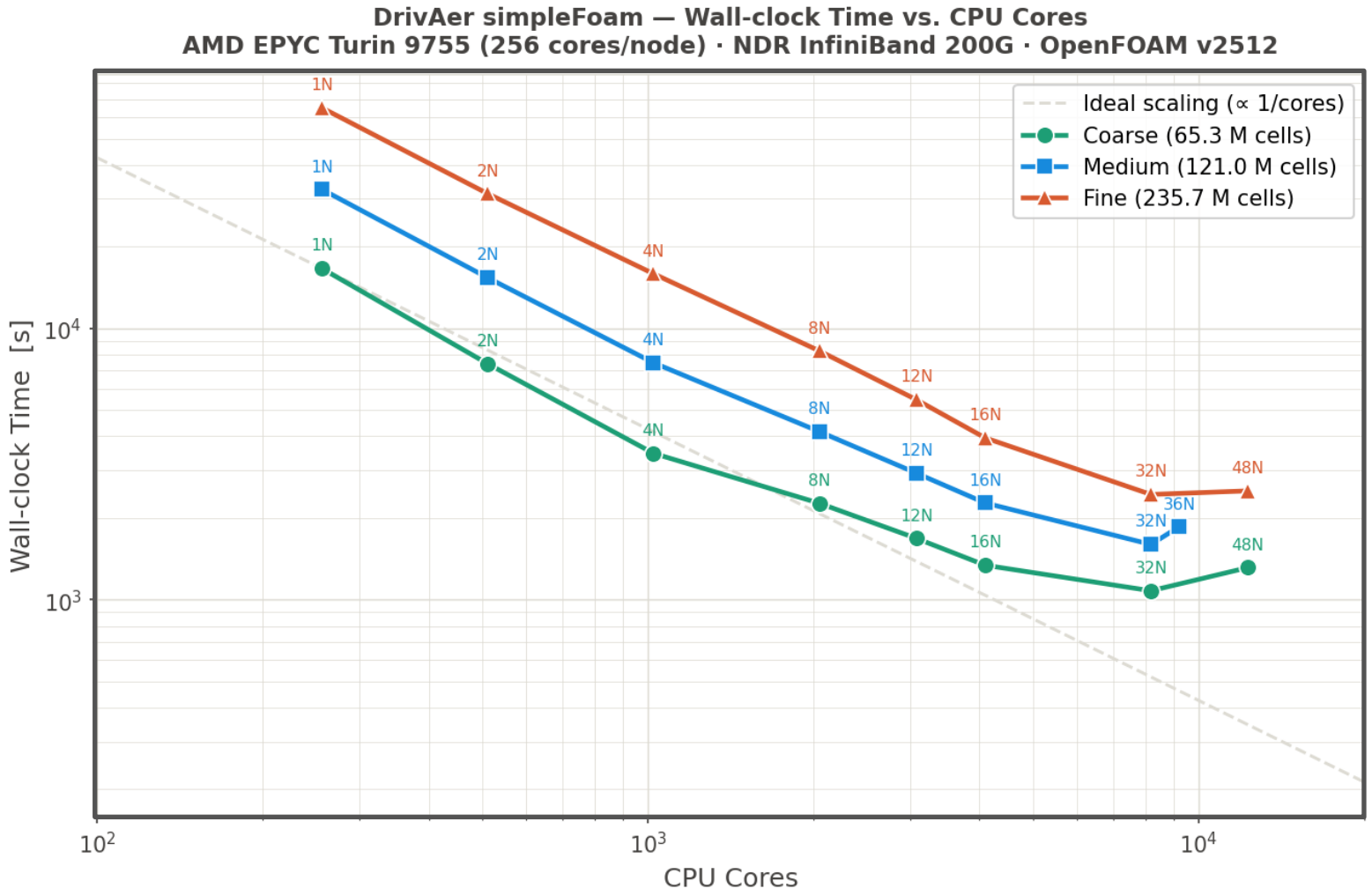
- Standard Case Setup – only change: number of partitions

	number of nodes used (256 cores each)							
coarse	1	2	4	8	12	16	32	48
medium	1	2	4	8	12	16	32	48
fine	1	2	4	8	12	16	32	48

48 nodes = 12 288 cores

- Our motivation:
  - How fast is our HPC in comparison to others?
  - How much faster compared to our „older“ cluster?
  - Learn from the community about scaling tests, bottlenecks, optimisation.
  - Exchange with others how to efficiently use OpenFOAM on HPCs.

Wall Clock Time



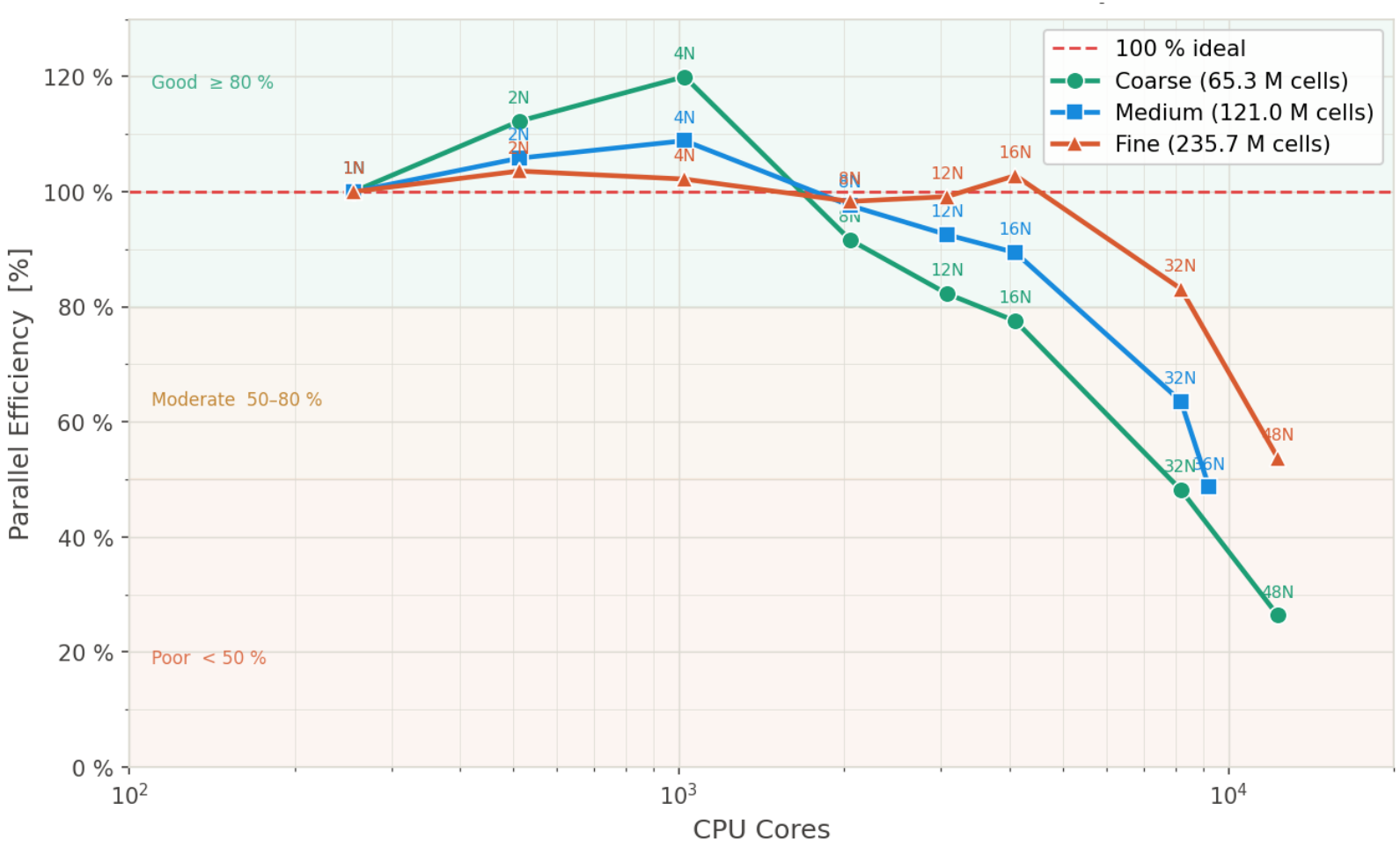
Speedup up to
32 nodes = 8 192 Cores
Total wall clock time gets shorter.

Shortest wall-clock time:
1078 s for the coarse mesh

WCT gets longer from 32 to 48.
Potential reasons:

- inter-switch traffic
  (40 ports/switch)
- Communication overhead

Parallel Efficiency



$$Parallel\ Efficiency = \frac{t_{wall}(1\ node)}{t_{wall}\ (N\ nodes) x\ N}$$

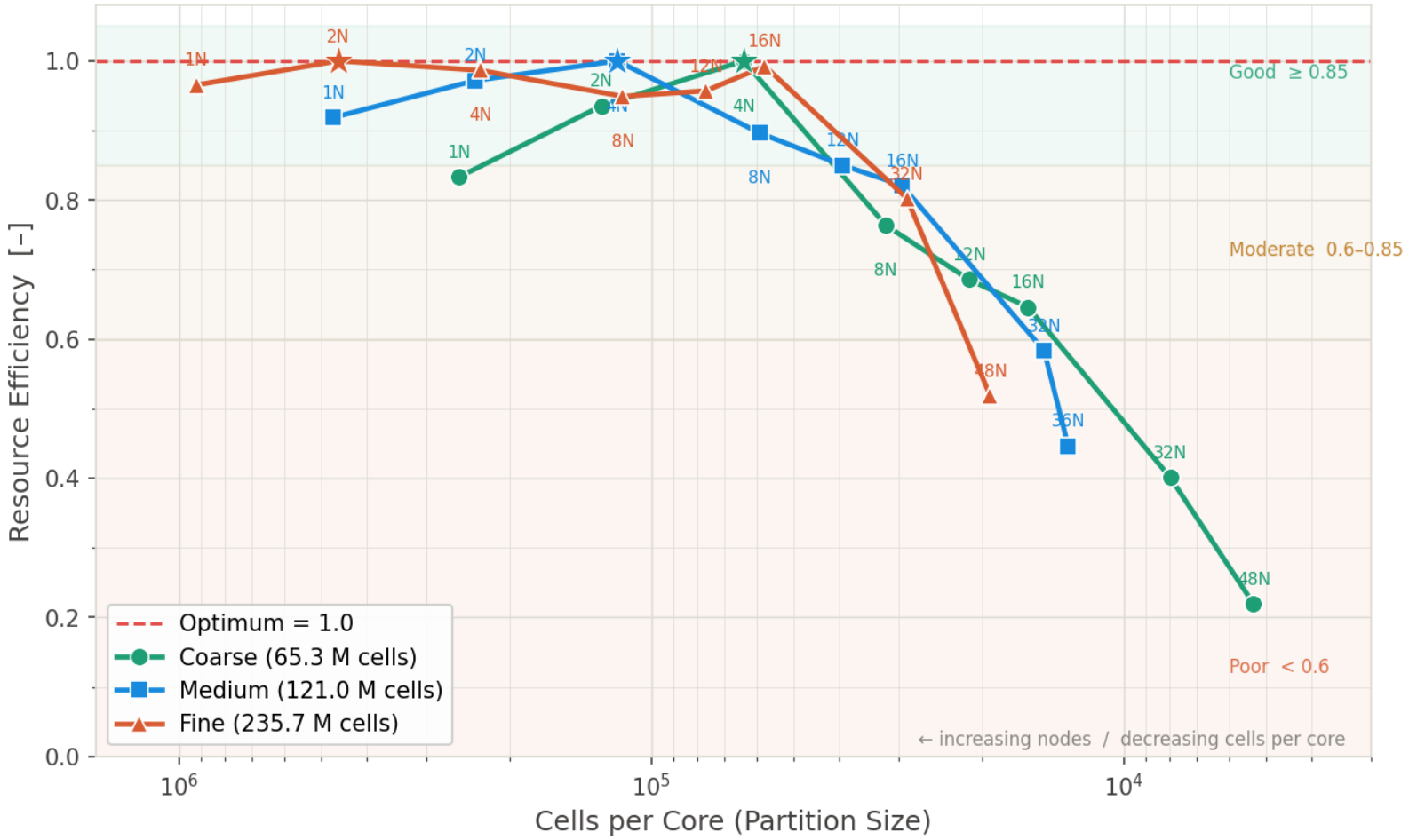
Parallel efficiency grows up to 4 nodes, then it decreases.

Up to 16 nodes the efficiency is acceptable.

With 32 nodes, the efficiency is only at 50 % for the coarse mesh.

⇒ ressources x32 more
time x15 less

Ressource usage efficiency

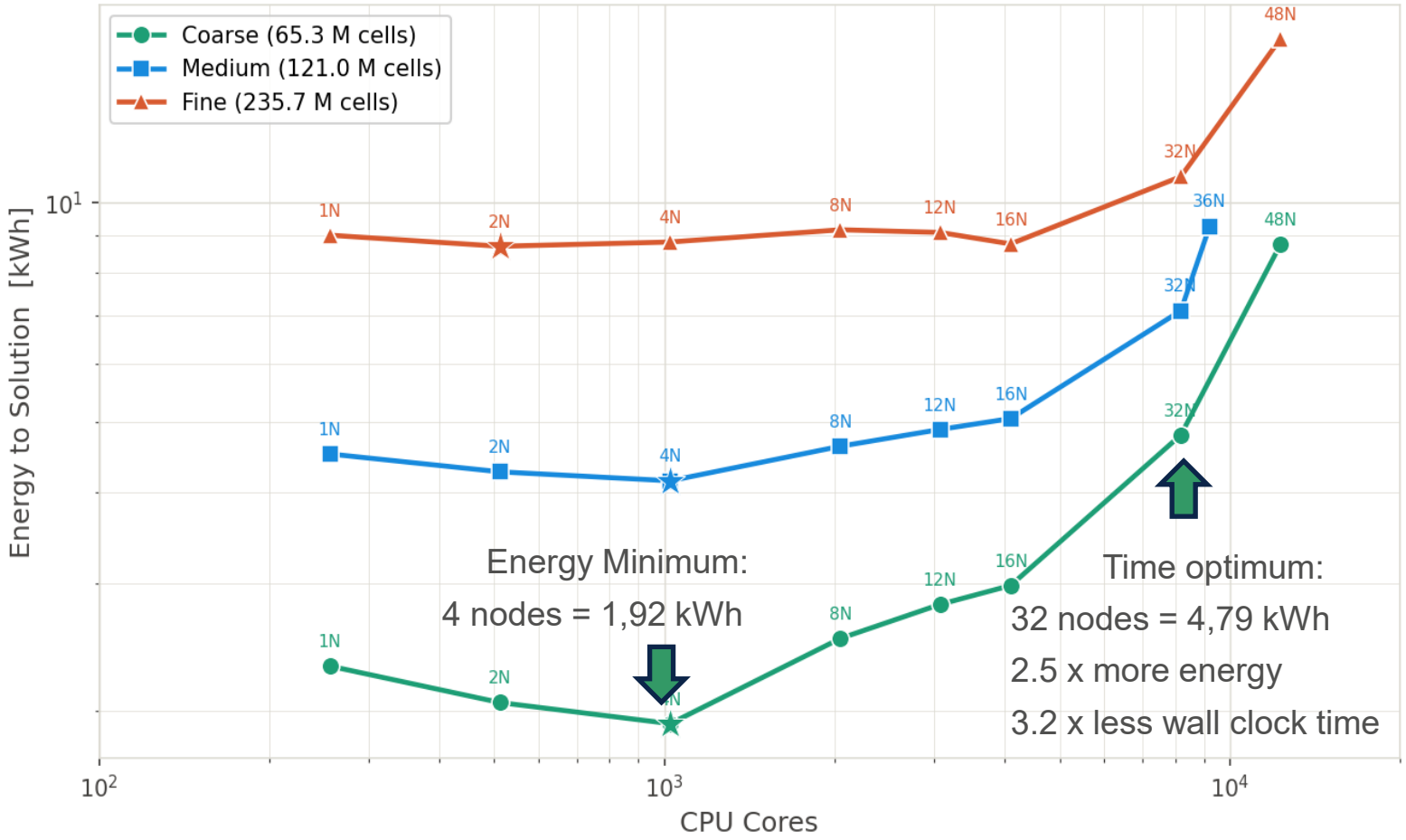


$$Ressource\ Efficiency = \frac{\min(t_{wall}\ x\ N_{cores})}{t_{wall}\ x\ N_{cores}}$$

The fine grid stays above 0.95 across a very wide range (2 nodes to 16 nodes), meaning it is remarkably tolerant of over-decomposition.

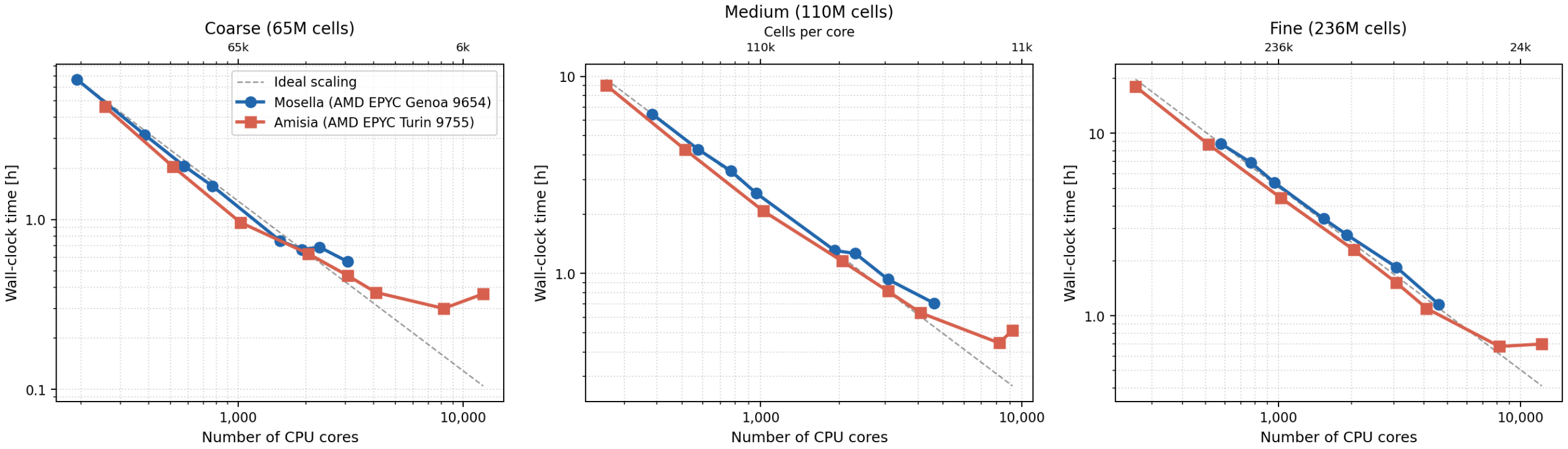
The coarse grid degrades much more steeply, confirming it should never be run beyond 16 nodes from a resource perspective.

Energy to solution



Over a large range of CPU cores, the energy to solution is not remarkably growing.

Comparison Amisia – Mosella (AMD Turin vs. AMD Genoa)



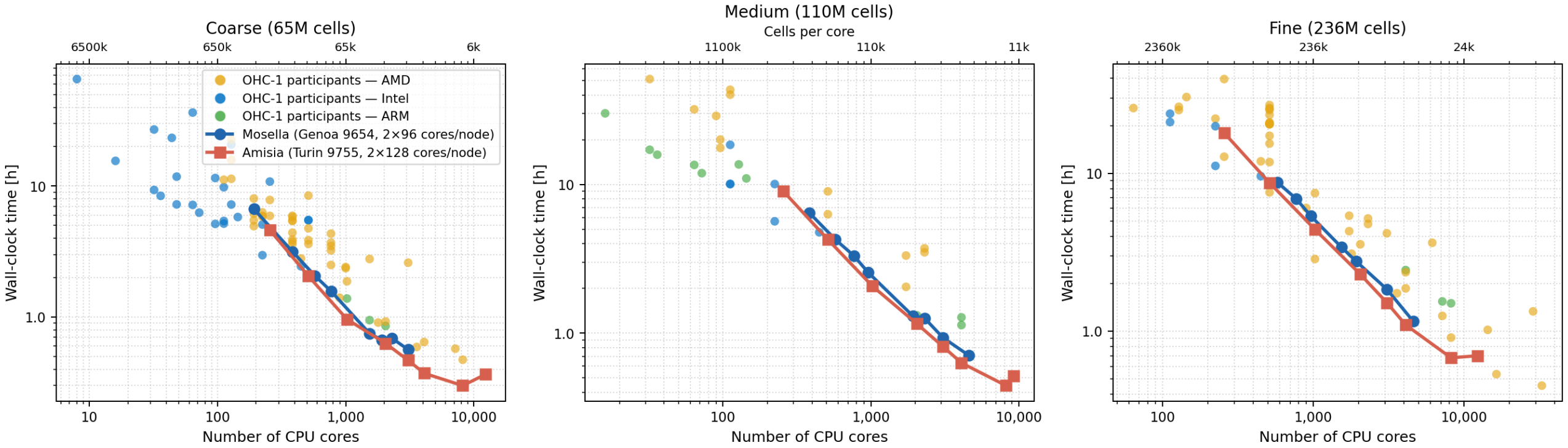
For the coarse mesh:

Cluster	Cores/node	FVOPS/node
Mosella	192	10,710,202
Amisia	256	15,544,297

Speedup of 1.15 -1.22 from Mosella to Amisia with same core count.

Reasons: - Fewer nodes -> less inter-node communication
- More bandwidth per node
- Zen 4 to Zen 5 architectural improvements

Comparison Amisia (red) and Mosella (blue) vs. Last years challenge OHC-1



Ideas to make the job faster (software track)

fvSolution_opt

```
p
{
    solver GAMG;
    smoother GaussSeidel;
    tolerance 0;
    relTol 0;
    maxIter 12;
    nCellsInCoarsestLevel 50;
    coarsestLevelCorr
    {   solver PCG;
        preconditioner DIC;
        tolerance 1e-4;
        relTol 0.01;
        minIter 1; // 35;
        maxIter 25; //35;
    }
}
```

Added changes

Ideas to make the job faster (software track)

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Improvement for 32 nodes:
from **0.270 s to 0.255 s** AvgWallClockTime

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Improvement for 32 nodes:
from **0.270 s to 0.255 s** AvgWallClockTime

decomposeParDict_opt

```
method              multiLevel;
multiLevelCoeffs
{
    nodes
    {
        numberOfSubdomains  32;
        method              scotch;
    }
    cores
    {
        numberOfSubdomains  256;
        method              scotch;
    }
}
```

Ideas to make the job faster (software track)

fvSolution_opt

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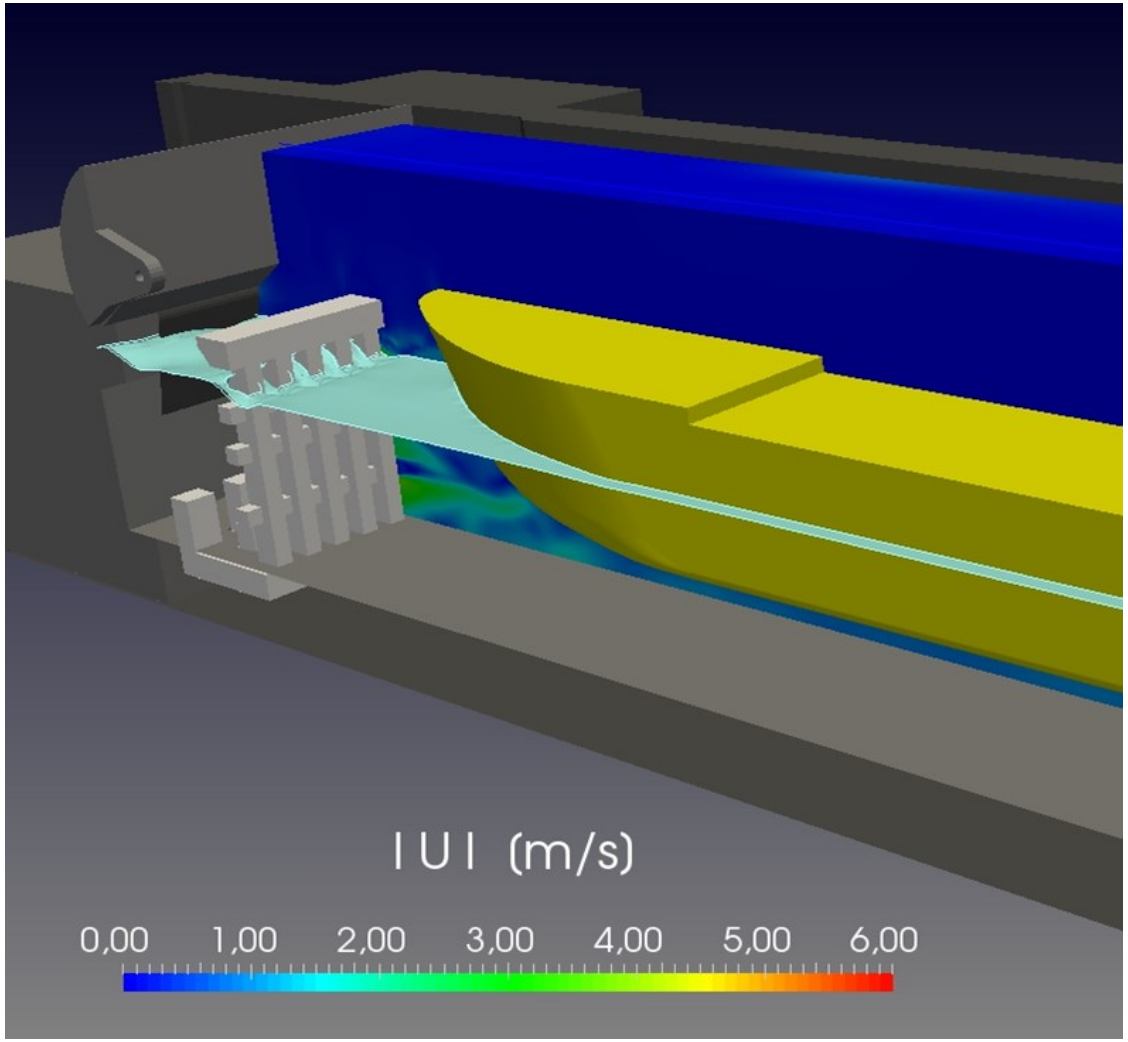
decomposeParDict_opt

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    cores
    {
        numberOfSubdomains  256;
        method              scotch;
    }
}
```

fvSolution_opt + decomposeParDict_opt:
Improvement for 32 nodes: from **0.255 s to 0.139 s** AvgWallClockTime

Conclusions

- Our hardware performs satisfactory
- Improvement of ~20% between AMD Genoa and AMD Turin
- Our best practice in our daily work: find optimum for efficient calculation, then run case study.
- Typically free-surface simulations (interFoam) with long runtime compared to decomposition time.
  Typical meshes ~5-50 Mio cells.
- We see that optimisation is strongly dependent on the grid quality and grid





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